

Inference for logistic regression models

Recap: hypothesis testing

Model:

$$Admit_i \sim Bernoulli(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 GPA_i$$

Research question: is there a relationship between GPA and acceptance to a graduate program?

Hypotheses: $H_0 : \beta_1 = 0$ $H_A : \beta_1 \neq 0$

Option 1: Wald test (aka z-test)

```
m1 <- glm(admit ~ gpa, family = binomial, data = grad)
summary(m1)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.357587	1.0353170	-4.20894	2.565715e-05
gpa	1.051109	0.2988694	3.51695	4.365354e-04

$$z = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} = 3.517$$

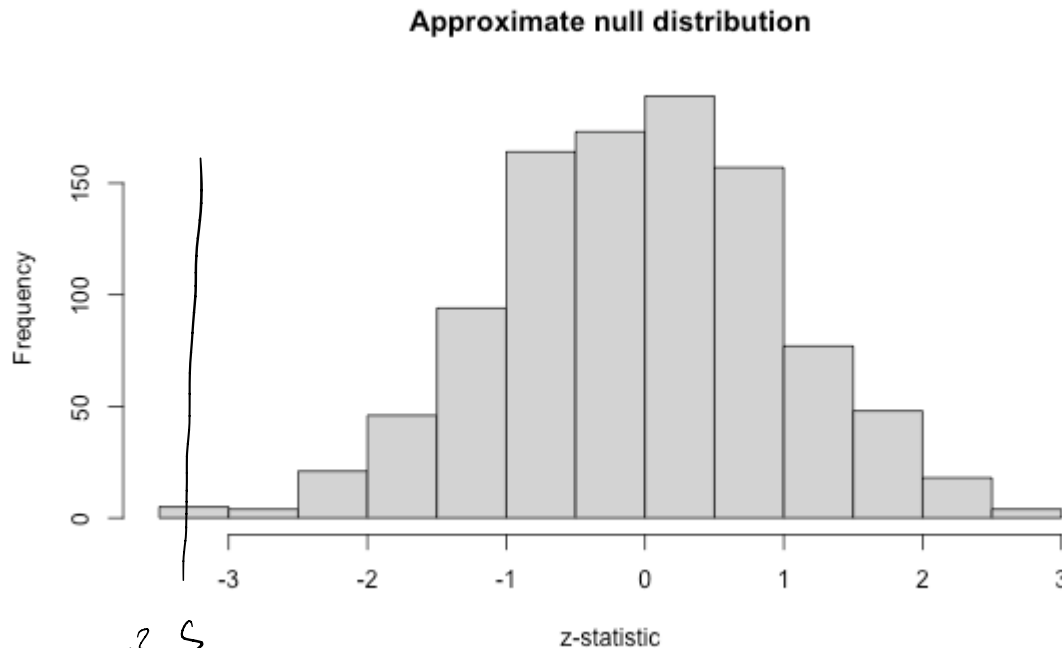
observed slope (1.051) is 3.5
standard deviations away from
value hypothesized under H_0 (0)

Is this an "unusual" z-statistic, if H_0 is true?

68 - 95 - 99.7 rule
1SD 2SD 3SD

Option 1: Wald test (aka z-test)

Approximate null distribution:



test statistics
calculated when
we forced H_0 to
be true (by
permuting)

← Standard normal
distribution
 $N(0, 1)$

- approximately normal
- centered at 0
- standard deviation ≈ 1

Option 1: Wald test (aka z-test)

Observed test statistic: $z = 3.517$

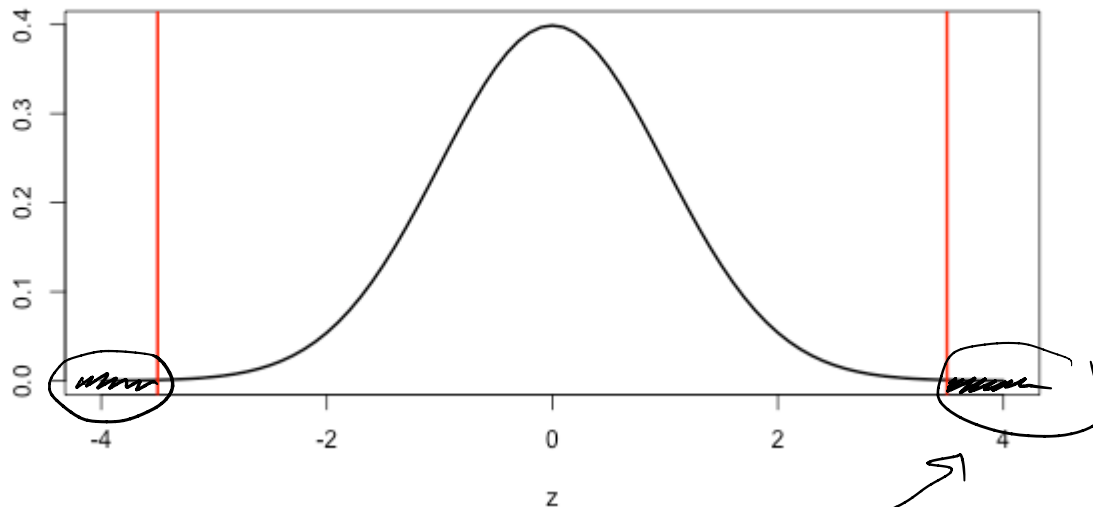
$$H_0: \beta_1 = 0$$

$$H_A: \beta_1 \neq 0$$

$\Rightarrow$ two-tailed test

$\beta_1 > 0$ right tail

$\beta_1 < 0$ left tail



```
2*pnorm(abs(3.517), lower.tail=F)
```

look at right tail

```
## [1] 0.0004364538
```

Option 1: Wald test (aka z-test)

```
2*pnorm(abs(3.517), lower.tail=F)
```

```
## [1] 0.0004364538
```

```
summary(m1)$coefficients
```

##		Estimate	Std. Error	z value	Pr(> z)
##	(Intercept)	-4.357587	1.0353170	-4.20894	2.565715e-05
##	gpa	1.051109	0.2988694	3.51695	4.365354e-04

Recap: hypothesis testing

full model: under H_A

reduced model: under H_0

Full model:

$$Admit_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 GPA_i$$

Wish to test $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$.

What is my **reduced model** for these hypotheses?

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = \beta_0$$

Recap: hypothesis testing

Full model:

$$Admit_i \sim Bernoulli(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 GPA_i$$

Wish to test $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$.

Reduced model:

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0$$

Activity

Work on the activity (handout), then we will discuss as a class.

Activity

- null deviance was same for both models (500)
- residual deviance for intercept-only model is also 500
- residual deviance for full model is lower

Fitted reduced model:

```
glm(admit ~ 1, family = binomial, data = grad_app)
```

```
##
```

```
## Call: glm(formula = admit ~ 1, family = binomial, data
```

```
##
```

```
## Coefficients:
```

```
## (Intercept)
```

```
## -0.7653
```

```
##
```

```
## Degrees of Freedom: 399 Total (i.e. Null); 399 Residual
```

```
## Null Deviance: 500
```

```
## Residual Deviance: 500 AIC: 502
```

null deviance: deviance for
intercept-only model

What do you notice about the "Null" and "Residual" deviance?

Activity

```
glm(admit ~ gpa, family = binomial, data = grad_app)
```

```
##  
## Call:  glm(formula = admit ~ gpa, family = binomial, data = grad_app)  
##  
## Coefficients:  
## (Intercept)          gpa  
##      -4.358         1.051  
##  
## Degrees of Freedom: 399 Total (i.e. Null);  398 Residual  
## Null Deviance:      500  
## Residual Deviance: 487      AIC: 491
```

What is the drop-in-deviance statistic for the test $H_0 : \beta_1 = 0$
vs. $H_A : \beta_1 \neq 0$?

$$G = 500 - 487 = 13$$

Activity

```
##  
## Call:  glm(formula = admit ~ gpa, family = binomial, data = dat)  
##  
## Coefficients:  
## (Intercept)          gpa  
##      -4.358          1.051  
##  
## Degrees of Freedom: 399 Total (i.e. Null);  398 Residual  
## Null Deviance:      500  
## Residual Deviance: 487      AIC: 491
```

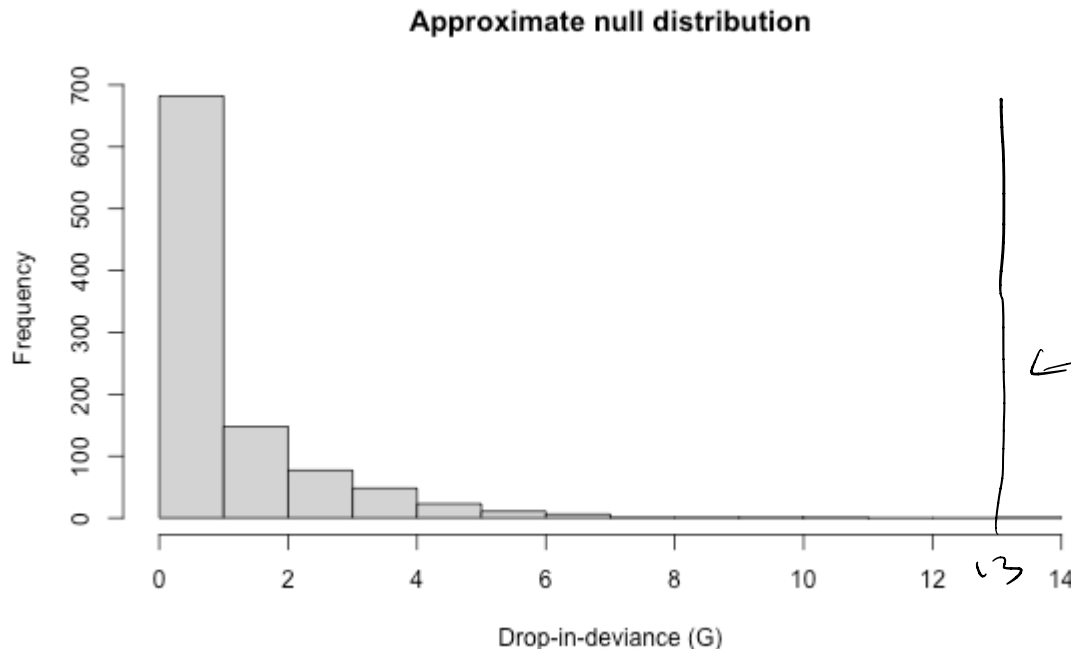
$$G = \text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}} = 13$$

How do we know if this test statistic is "unusual" under the null hypothesis?

Compare to null distribution of G
(distribution of G if H_0 were true)

Activity

Approximate null distribution:

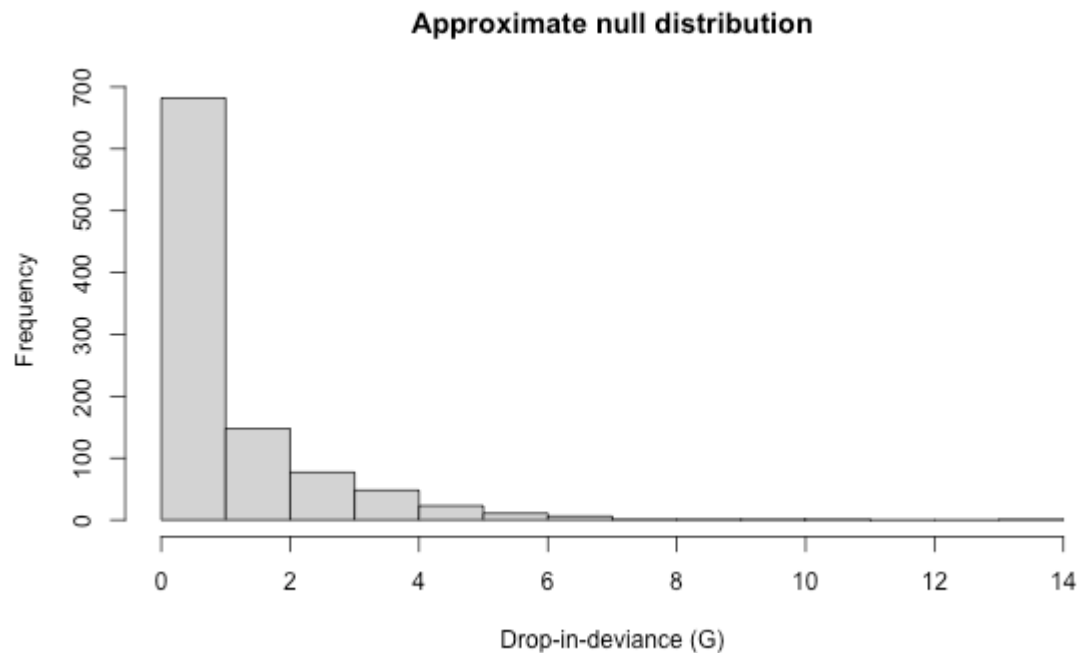


Look at distribution of G when we make H_0 true (by permuting)

← very unusual if H_0 were true!

Our observed test statistic is $G = 13$. Do you think that the null hypothesis is true?

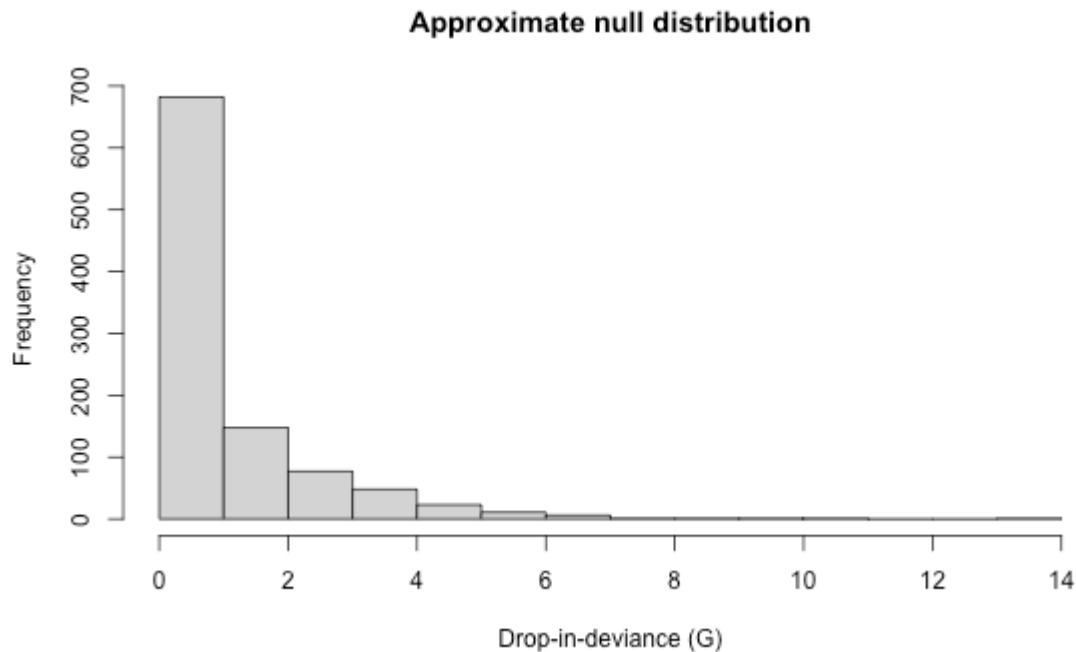
Calculating a p-value



p-value = $P(G > \text{observed value} \mid H_0 \text{ true})$

Need a distribution to calculate p-value!

Class activity

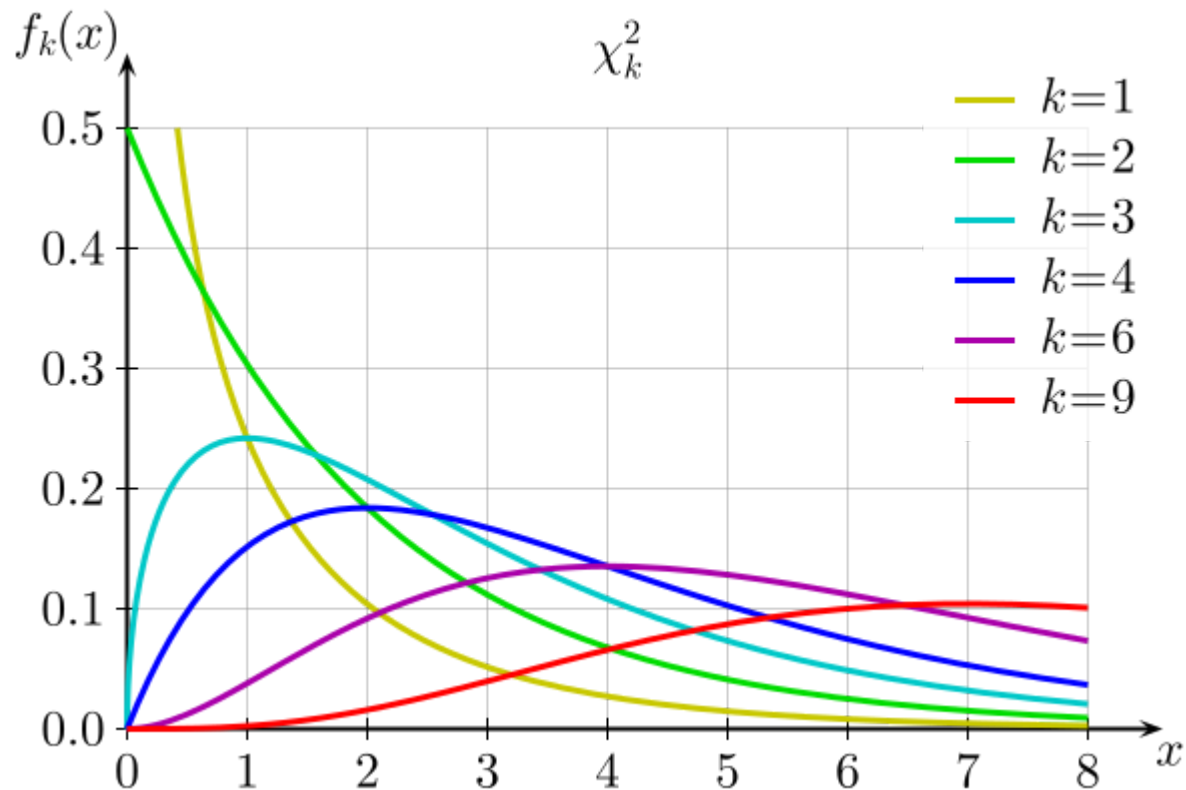


Why is the drop-in-deviance always non-negative?

can never increase residual deviance by adding more parameters

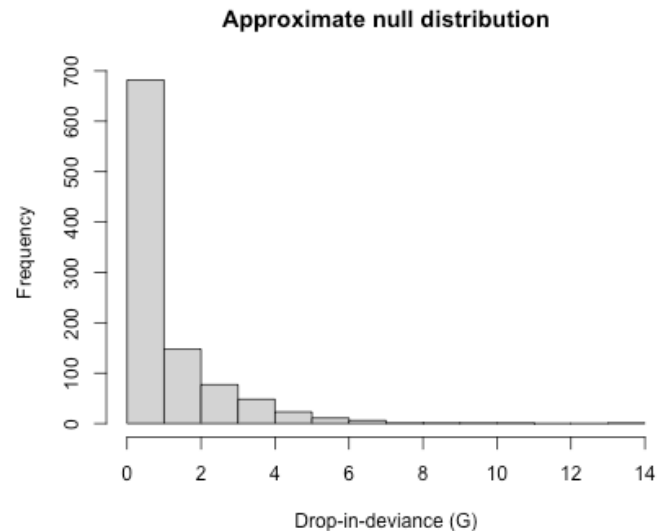
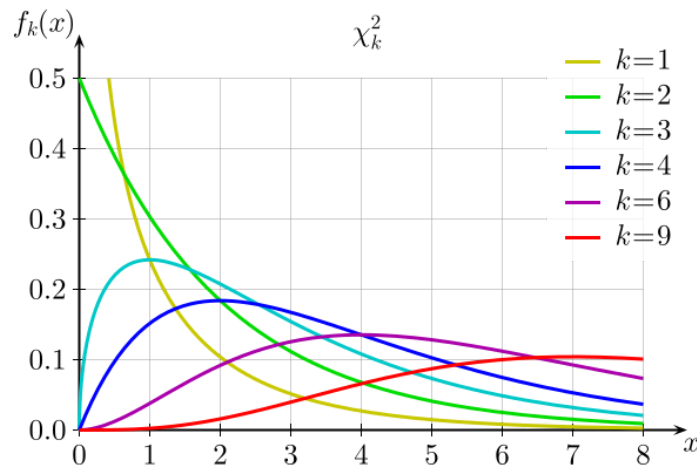
Class activity

Under H_0 , $G \sim \chi_k^2$



Class activity

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$



Under H_0 , $G \sim \chi_k^2$. Which value of k seems appropriate for the null distribution?

Class activity

- + Mean of χ_k^2 is k
- + Variance of χ_k^2 is $2k$

```
mean(null_statistics)
```

```
## [1] 1.020683
```

```
var(null_statistics)
```

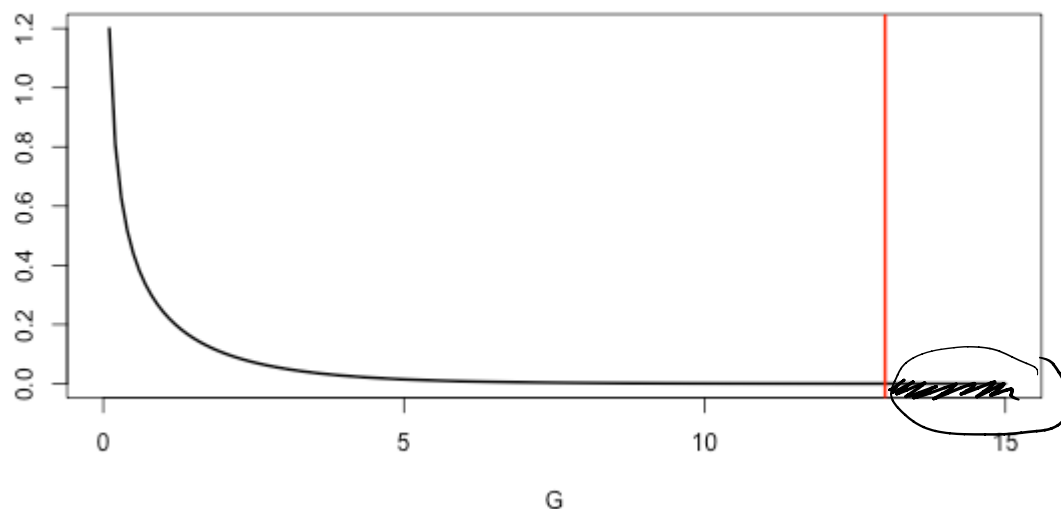
```
## [1] 2.00598
```

$\Rightarrow k=1$

Calculating a p-value

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0 \quad G = \text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}} = 13$$

Under $H_0, G \sim \chi_1^2$



```
pchisq(13, df=1, lower.tail=F)
```

```
## [1] 0.000311491
```

Drop-in-deviance test (aka likelihood ratio test)

Goal: Compare nested models (full and reduced models)

Steps:

- + Calculate deviance for full and reduced models
- + Test statistic: $G = \text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}}$
- + Null distribution: under H_0 , $G \sim \chi^2_q$

$\uparrow$
 $q = \#$ parameters tested

$$H_0: \beta_1 = 0 \quad H_A: \beta_1 \neq 0$$

$$q = 1$$

$$H_0: \beta_1 = \beta_2 = 0$$

$$q = 2$$

Wald test vs. drop-in-deviance test

Wald test (z-test)

- + like t-tests
- + test a *single* parameter
- + some example hypotheses:
 - + $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$
 - + $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 > 0$
- + null distribution: $N(0, 1)$

Drop-in-deviance test (likelihood ratio test)

- + like nested F-tests
- + test one or more parameters
- + some example hypotheses:
 - + $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$
 - + $H_0 : \beta_1 = \beta_2 = 0$ vs H_A : at least one of $\beta_1, \beta_2 \neq 0$
- + null distribution: χ_q^2

Activity

Work on the activity (handout), then we will discuss as a class.

Activity

$$Admit_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 GPA_i + \beta_2 GRE_i + \beta_3 Rank2_i + \beta_4 Rank3_i + \beta_5 Rank4_i$$

We want to test whether there is a relationship between Rank and grad school acceptance, after accounting for GPA and GRE. What are my null and alternative hypotheses?

Activity

$$Admit_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 GPA_i + \beta_2 GRE_i + \beta_3 Rank2_i + \beta_4 Rank3_i + \beta_5 Rank4_i$$

$$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_A : \text{at least one of } \beta_3, \beta_4, \beta_5 \neq 0$$

What is the reduced model?

Activity

```
##  
## Call: glm(formula = admit ~ gpa + gre + factor(rank), family = binomial,  
##      data = grad_app)  
##  
## Coefficients:  
##      (Intercept)              gpa              gre factor(rank)2 factor(rank)3  
##      -3.989979         0.804038         0.002264         -0.675443         -1.340204  
## factor(rank)4  
##      -1.551464  
##  
## Degrees of Freedom: 399 Total (i.e. Null);  394 Residual  
## Null Deviance:          500  
## Residual Deviance: 458.5      AIC: 470.5
```

```
##  
## Call: glm(formula = admit ~ gpa + gre, family = binomial, data = grad_app)  
##  
## Coefficients:  
##      (Intercept)              gpa              gre  
##      -4.949378         0.754687         0.002691  
##  
## Degrees of Freedom: 399 Total (i.e. Null);  397 Residual  
## Null Deviance:          500  
## Residual Deviance: 480.3      AIC: 486.3
```

Activity

$$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_A : \text{at least one of } \beta_3, \beta_4, \beta_5 \neq 0$$

Drop-in-deviance test: $G = 21.8$

What is the distribution of G under the null hypothesis?

Activity

$$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_A : \text{at least one of } \beta_3, \beta_4, \beta_5 \neq 0$$

Drop-in-deviance test: $G = 21.8$

Under H_0 , $G \sim \chi_3^2$

```
pchisq(21.8, df=3, lower.tail=F)
```

```
## [1] 7.178958e-05
```